

Interdisciplinary Guide — T1/T4 & InterIA

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Abstract

This 2–3 page guide explains the T1/T4 plan, translation between disciplines, and how to read the proof without switching hats. InterIA artifacts are separate.

1. Reading routes (personas)

1 Reading routes (by specialty)

NT. T1/T2 → T3; T4 minimal.

AO. T2 → T1/T2 → T3; T4 as spectral reading.

Signal. Tapers, bandlimit; T2 as frame conditioning; T3 as matched-filter probe.

Phys. Rigidity (T2) + localized measurement (T3); HP compatible.

Repro. See InterIA (separate).

Formal. Lemmas: Schur/Gershgorin, PW density, Archimedean, absorption.

2. Cross-dictionary

2 Notations and preliminaries

NT	Cross reading
Band $[-A, A]$	Bandwidth (signal)
Cos^2 taper	Apodization (plateau/transition)
Ξ_A	Non-uniform sampling
$\log p/p^{k/2}$	p -adic energy weighting
$C_{\text{fr}}^{(w)} \geq 1 - \Theta_w$	Frame conditioning (AO)
Wave-packet $\hat{\phi}_T$	Matched filter (signal/phys)
Explicit formula	Source-spectrum coupling

3. Storyboard

3 Storyboard T1/T4

T1/T2: set the arena (PW, explicit, Archimedean).

T2: gap via locality + weights.

T3: packet → contradiction if off-line zero.

T4: HP-compatible reading (no postulate).

4. Bridges

4 Bridges boxes

(AO) $G^{(w)}$ Gram; $I + R$ with $\|R\| < 1$ gives lower boundedness.

(Signal) Band $[-A, A]$, apodized taper, frame $\approx$ robust reconstruction.

(Phys) T2 = spectral rigidity; T3 = localized measurement.

(Repro) Interval certificates; proof does not depend on numerics.

5. Micro-example (T2)

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$A = 4$; $p \leq e^4$.

Few neighbors (locality), decay $p^{-(k-k_\alpha)/4}$.

Indicative: $\Theta_w(A) \approx 0.63 \Rightarrow C_{\text{fr}}^{(w)} \approx 0.37 > 0$.

6. FAQ

6 FAQ (by specialty)

NT: why $p^{-k/2}$? — damp prime powers.

AO: why Schur? — direct control of $\|R\|$.

Signal: why $\cos^2$? — bounded slope, simple calculus.

Phys: HP? — reading only.

Repro: numbers? — certificates, not part of the proof.

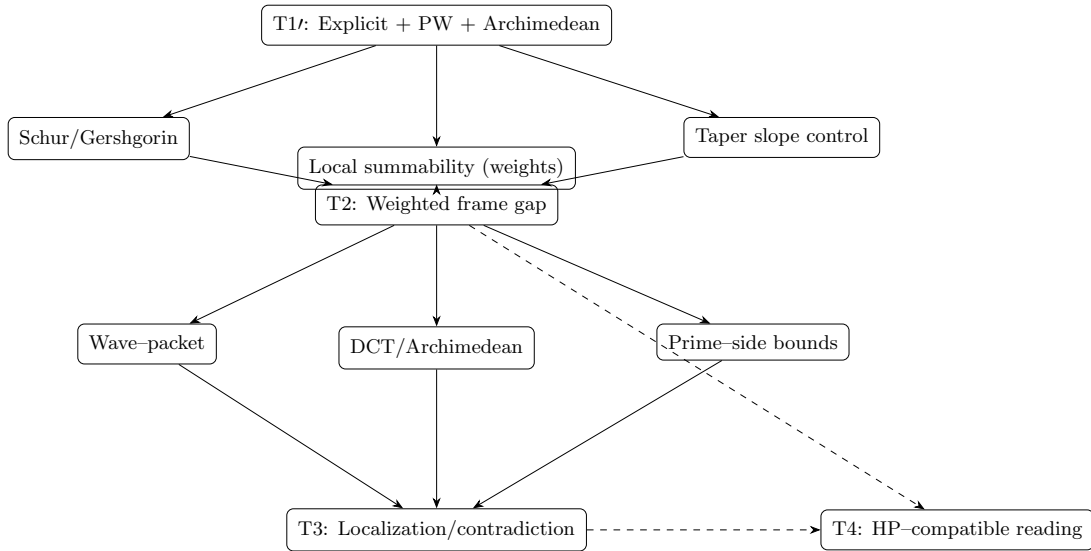


Figure 1: Dependency map: $T1 \rightarrow T2 \rightarrow T3$; T4 is HP-compatible reading.